WEYL ASYMPTOTICS AND HEAT KERNELS. TAUBERIAN THEORY, INTEGRAL KERNELS, HILBERT SCHMIDT, TRACE CLASS, AND MERCER'S THEOREM.

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ABSTRACT. For kernels, trace class, Hilbert Schmidt operators and Mercer's theorem, see [Wikb], [Lax02, p329].

CONTENTS

1. Dirichlet and Neumann Problems

We recall some standard material.

Fix $f \in L^2(\Omega)$. Then $u \in H_0^1(\Omega)$ is defined to be a weak solution of the *Dirichlet problem*

$$\Delta u = f \text{ in } \Omega, \quad u = 0 \text{ on } \partial\Omega, \tag{1}$$

if

$$\int Du\, Dv = -\int_\Omega fv \quad \forall v \in H_0^1(\Omega). \tag{2}$$

If u and f are smooth, it follows from integration by parts and the boundary condition, that this is equivalent to the classical notion in (1).

Since

$$\left| \int_\Omega fv \right| \le \|f\|_{L^2} \|v\|_{L^2} \le c\|f\|_{L^2} \|v\|_{H_0^1},$$

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it follows that $v \mapsto \int fv$ is a bounded linear functional on H_0^1, and so by the Riesz representation theorem *there exists a unique solution of* (2) *and hence of* (1).

We write $u = \Delta^{-1} f$, where $\Delta^{-1} : L^2 \to H_0^1 \hookrightarrow L^2$ is the "inverse of the Dirichlet Laplacian".

Since $H_0^1 \hookrightarrow L^2$ is compact it follows $\Delta^{-1} : L^2 \to L^2$ is bounded and compact. From (2) it follows that Δ^{-1} is symmetric w.r.t. the L^2 inner product. From (2) and Poincare's inequality it follows that Δ^{-1} is positive definite.

It follows from the spectral theory for compact bounded self adjoint operators, applied to Δ^{-1}, that there exist eigenvalues $0 < \lambda_1 \le \lambda_2 \le \lambda_3 \cdots$ and corresponding eigenfunctions $\phi_n \in H_0^1 \subset L^2$ for Δ. That is

$$\Delta\, \phi_n = -\lambda_n \phi_n.$$

It also follows that (after normalisation), (ϕ_n) is a complete orthonormal basis for L^2.

It follows from standard regularity theory [Eva98, Remark on p 335] that the ϕ_n are C^∞ if $\partial\Omega$ is C^∞. In particular, $\phi_n = 0$ on $\partial\Omega$ since $\phi_n \in H_0^1$.

It follows from the variational characterisation of eigenvalues that $\lambda_1 < \lambda_2$, see [Eva98, Theorem 2, p 336].

Using (2),

$$\int D\phi_n D\phi_m = \lambda_n \int \phi_n \phi_m = \delta_{nm}, \qquad \int |D\phi_n|^2 \ge c \int |\phi_n|^2 > 0,$$

Hence $(\lambda_n^{-1/2} \phi_n)$ is an orthonormal basis for H_0^1.

Fix $f \in L^2(\Omega)$. Then $h \in H_1(\Omega)$ is said to be a weak solution of the *Neumann problem*

$$\Delta\, u = f \text{ in } \Omega, \quad \frac{\partial u}{\partial \nu} = 0 \text{ on } \partial\Omega, \tag{3}$$

if

$$\int_\Omega Du\, Dv = -\int_\Omega fv \quad \forall v \in H^1(\Omega). \tag{4}$$

(Note $v \in H^1(\Omega)$, no boundary conditions are imposed.) If u and f are smooth, it follows from integration by parts, standard regularity theory and the boundary condition, that this is equivalent to the classical notion in (3).

If there *is* a solution of (4), setting $v = 1$ implies $\int f = 0$. Under this assumption and working in the space $\text{Ł}^2 := \{f \in L^2 : \int f = 0\}$, it follows as before that *there exists a unique solution* $u \in L^2$ *of* (4) *and hence of* (3).

We write $u = \Delta^{-1} f$, where now $\Delta^{-1} : \text{Ł}^2 \to H^1 \cap \{f : \int f = 0\} \hookrightarrow \text{Ł}^2$ is the "inverse of the *Neumann* Laplacian". Again similarly to before, $\Delta^{-1} : \text{Ł}^2 \to \text{Ł}^2$ is compact, symmetric w.r.t. the L^2 inner product, and positive definite on Ł^2.

Also, there exist eigenvalues $0 < \gamma_1 \le \gamma_2 \le \gamma_3 \cdots$ and corresponding eigenfunctions $\psi_n \in H^1 \cap \{f : \int f = 0\} \subset \text{Ł}^2$. That is

$$\Delta\, \psi_n = -\gamma_n \psi_n.$$

(The constant eigenfunctions are not in Ł^2 except for the zero function.) After normalisation, (ψ_n) is a complete orthonormal basis for Ł^2.

It follows from standard regularity theory that the ψ_n are C^∞ if $\partial\Omega$ is C^∞. In particular, $\dfrac{\partial u}{\partial \nu} = 0$ on $\partial\Omega$.

Using (4),

$$\int D\psi_n D\psi_m = \gamma_n \int \psi_n \psi_m = \delta_{nm}, \qquad \int |D\psi_n|^2 \ge c \int |\psi_n|^2 > 0.$$

Hence $(\gamma_n^{-1/2} \psi_n)$ is an orthonormal basis for $H^1 \cap \{f : \int f = 0\}$.

Example 1.1.

(1) The usual Fourier orthonormal basis for $L^2(-\pi,\pi)$ is

$$\frac{1}{\sqrt{2\pi}},\ \frac{1}{\sqrt{\pi}}\cos nx,\ \frac{1}{\sqrt{\pi}}\sin nx \quad (n\geq 1).$$

They are the pointwise solutions of $\Delta\,\phi=\lambda\phi$ on $[-\pi.\pi]$ for some λ, i.e. the trigonometric functions, which satisfy $\phi(-\pi)=\phi(\pi)$.

They is obtained by working with the Laplacian on the compact manifold S^1 essentially as before but with no boundary restrictions. Or equivalently by working with the space of 2π-periodic functions.

(2) Testing $\cos\alpha x$ and $\sin\alpha x$ for the Dirichlet boundary condition gives the following orthonormal basis for $L^2(-\pi,\pi)$:

$$\frac{1}{\sqrt{\pi}}\cos(n-1/2)x,\ \frac{1}{\sqrt{\pi}}\sin nx \quad (n\geq 1).$$

They are the pointwise solutions of $\Delta\,\phi=\lambda\phi$ on $[-\pi.\pi]$, i.e. the trigonometric functions, which satisfy $\phi(-\pi)=\phi(\pi)=0$.

(3) Testing $\cos\alpha x$ and $\sin\alpha x$ for the Neumann boundary condition gives the following orthonormal basis for $Ł^2(-\pi,\pi)$:

$$\frac{1}{\sqrt{\pi}}\cos nx,\ \frac{1}{\sqrt{\pi}}\sin(n-1/2)x \quad (n\geq 1).$$

Adding the constant function $1/\sqrt{2\pi}$ gives the orthonormal basis

$$\frac{1}{\sqrt{2\pi}},\ \frac{1}{\sqrt{\pi}}\cos nx,\ \frac{1}{\sqrt{\pi}}\sin(n-1/2)x \quad (n\geq 1).$$

for $L^2(-\pi,\pi)$. They are the pointwise solutions of $\Delta\,\phi=\lambda\phi$ on $[-\pi.\pi]$, i.e. the trigonometric functions, which satisfy $\phi'(-\pi)=\phi'(\pi)=0$.

2. Heat Equation Basics in $\mathbb{R}^n$

See [Bas08]. *For a very nice treatment of some of the following with* $\Omega\subset\mathbb{R}^d$ *see* [Dod81].

2.1. **Heat Equation.** Consider the heat equation either on a bounded domain $\Omega\subset\mathbb{R}^n$, in particular $(0,L)\subset\mathbb{R}$, or a compact Riemannian manifold (M,g).

For Ω,

$$\Delta=\frac{\partial^2}{\partial x_1^2}+\cdots+\frac{\partial^2}{\partial x_1^2}.$$

For M,

$$\Delta=\frac{1}{\sqrt{g}}\frac{\partial}{\partial x_i}\left(g^{ij}\sqrt{g}\frac{\partial}{\partial x_j}\right).$$

Thus we take the usual negative laplacian in either case.

The *heat equation* for $u=u(t,x)$ is

$$\begin{aligned}\partial_t u&=\Delta\,u \quad t>0,\ x\in\Omega \text{ or } M,\\ u(0,x)&=f(x)\quad \text{(initial condition)}.\end{aligned}\tag{5}$$

In the case of Ω for the Dirichlet problem, zero boundary condition $u(x,t)=0$ for $x\in\partial\Omega$. For M and for the Neumann problem on Ω (see Section 1) no boundary conditions are imposed, except that one can assume $\int u(0,x)\,dx=0$ — this condition is preserved under the flow.

2.2. **Fundamental Solution.** The *fundamental solution* is written $H=H(t,x,y)$ and is uniquely characterised by[1]

$$\partial_t H(t,x,y)=\Delta_x\,H(t,x,y),\ t>0,\ x,y\in\Omega\,(M),\tag{6}$$

$$H(0,x,y)=\delta_y(x),\quad (\text{i.e. } H(t,\cdot,y)\to\delta_y \text{ as } t\to 0).\tag{7}$$

In the case of Ω, $H(t,x,y)$ is also required to satisfy the relevant boundary condition, which in the Dirichlet case is

$$H(t,x,y)=0 \text{ if } t>0,\ y\in\Omega,\ x\in\partial\Omega.\tag{8}$$

[1]See [Eva98, pp46, 47] for a discussion of this in the case $\Omega=\mathbb{R}^n$.

The motivation is that $H(t,x,y)$ is the solution at time t and place x with initial condition δ_y (a "unit pulse" at y) applied at time 0.

It follows from regularity theory that H is smooth for $t > 0$.

It follows from Proposition 2.1 that $H(t,x,y) = H(t,y,x)$.

It follows from (9) that

$$H(t+s,x,y) = \int H(t,x,z)\,H(s,z,y)\,dz.$$

The condition (7) says

$$\lim_{t\to 0+} \int_\Omega H(t,x,y)\,f(y)\,dy = f(x),$$

uniformly for $f \in C(\overline{\Omega})$ and $f = 0$ on $\partial\Omega$.

Given an initial condition $f(x)$ it follows that

$$u(t,x,y) = \int H(t,x,y)\,f(y)\,dy. \tag{9}$$

Thus we can regard H as a linear operator from an appropriate space of functions over Ω (M) into a space of functions over Ω (M) for each $t > 0$, or into a space of functions over $\mathbb{R}^+ \times \Omega$ ($\mathbb{R}^+ \times M$).

Example: (See [Ful04].) The heat kernel for the interval $[0,L]$ is

$$H(t,x,y) = \frac{1}{\sqrt{4\pi t}} \sum_{n=-\infty}^{\infty} \left(e^{-(x-y+2nL)^2/4t} \pm e^{-(x+y+2nL)^2/4t} \right). \tag{10}$$

where "+" corresponds to the Neumann problem and "−" to the Dirichlet problem. This can be shown by the "method of images".

Note also that

$$\begin{aligned}\int_0^L H(t,x,x)\,dx &= \frac{1}{\sqrt{4\pi t}} \left(\int_0^L \left(1 \pm e^{-x^2/t} \pm e^{-(x-L)^2/t}\right) + O(t) \right) \\ &= \frac{L}{\sqrt{4\pi t}} \pm \frac{1}{2} + O(t^{1/2}).\end{aligned}$$

Thus from the short term asymptotics one can read off the length and the type of boundary condition.

The *probabilistic interpretation* of H is that for each $y \in \Omega\,(M)$ and $t > 0$, $H(t,x,y)$ is the probability density for a Brownian particle at y at time 0 to move to x at time t.

2.3. **Duhamel's Principle.** This allows us to solve the initial value problem (5) with a forcing term, that is to solve

$$\begin{aligned}\partial_t u(t,x) &= \Delta_x\, u(t,x) + g(t,x) \quad t > 0,\ x \in \Omega \text{ or } M, \\ u(0,x) &= f(x) \quad \text{(initial condition)}.\end{aligned} \tag{11}$$

The idea is to treat the contribution due to the forcing term up to time t as a "sum" (i.e. integral) of initial data terms of the form $g(s,y)\,ds$ in the time interval $[s, s+ds]$. Thus we conjecture the solution is

$$u(t,x) = \int_\Omega H(t,x,y)\,f(y)\,dy + \int_0^t \int_\Omega H(t-s,x,y)\,g(s,y)\,dy\,ds. \tag{12}$$

To see this formally we compute for $t > 0$ and smooth data that

$$\begin{aligned}\partial_t u(t,x) - \Delta_x\, u(t,x) &= \int_\Omega (\partial_t - \Delta_x) H(t,x,y)\,f(y)\,dy + \int_\Omega H(0,x,y)\,g(t,y)\,dy \\ &\quad + \int_0^t \int_\Omega (\partial_t - \Delta_x) H(t-s,x,y)\,g(s,y)\,dy\,ds \\ &= 0 + g(t,x) + 0 = g(t,x).\end{aligned} \tag{13}$$

The initial condition is satisfied as before.

This idea is also used in constructing a parametrix, see pages 9 ff.

2.4. **Semigroup Approach and Eignefunction Expansion for Heat Kernel.** (See also Section 3.) Thinking of the heat equation (5) as an ODE on the space $H_0^1(\Omega)$, *we write* $e^{t\,\Delta}$ *for the integral operator* H and $e^{t\,\Delta}f(x)$ for the solution to (5). This is motivated by the fact that the solution of the O.D.E.

$$\frac{d}{dt}x(t) = Lx, \quad x(0) = x_0,$$

where $x(t) \in \mathbb{R}^d$ and $L : \mathbb{R}^d \to \mathbb{R}^d$ is linear, is given by

$$x(t) = e^{tL}x_0 := \left(\sum_{n\geq 0}\frac{1}{n!}(tL)^n\right)x_0.$$

Let the eigenvalues and eigenfunctions of Δ be λ_n and ϕ_n for $n \geq 1$. They are the smooth pointwise solutions of $\Delta\,\phi = -\lambda\phi$, which in the case of Ω also satisfy the boundary conditions in (2) or (4).

Proposition 2.1. *The fundamental solution is given by the integral operator whose kernel is* $\sum_n e^{-t\lambda_n}\phi_n(x)\phi_n(y)$. *That is,*

$$e^{t\,\Delta} := H(t,x,y) = \sum_n e^{-t\lambda_n}\phi_n(x)\phi_n(y). \tag{14}$$

Proof.

$$\begin{aligned}(\partial_t - \Delta_x)H(t,x,y) &= (\partial_t - \Delta_x)\sum_n e^{-t\lambda_n}\phi_n(x)\phi_n(y)\\ &= \sum_n(-\lambda_n+\lambda_n)e^{-t\lambda_n}\phi_n(x)\phi_n(y) = 0,\end{aligned}$$

and

$$\begin{aligned}u(t,x) := \int H(t,x,y)f(y)\,dy &= \int\sum_n e^{-t\lambda_n}\phi_n(x)\phi_n(y)f(y)\,dy\\ &= \sum_n e^{-t\lambda_n}a_n\phi_n(x) \quad \text{where } a_n := (f,\phi_n)\\ &\to \sum_n a_n\phi_n(x) = f(x) \quad \text{in the } L^2 \text{ sense as } t\to 0,\end{aligned}$$

if for example $\sum_n \lambda_n^2 a_n^2 < \infty$, since then

$$\sum_n \left|e^{-t\lambda_n}-1\right|^2 a_n^2 \leq \sum_n t^2\lambda_n^2a_n^2 \to 0 \text{ as } t\to 0.$$

□

Note that the initial data $u(0,x) = f(x) = \sum_n a_n\phi_n(x)$ evolves according to

$$u(t,x) = \sum_n e^{-t\lambda_n}a_n\phi_n(x). \tag{15}$$

This justifies the notation $u(t,x) = e^{t\,\Delta}f(x)$.

Proposition 2.2. *For* $t > 0$, $e^{t\,\Delta}$ *is trace class and* $\operatorname{Tr} e^{t\,\Delta} = \sum_n e^{-t\lambda_n}$.

Proof. Note that the eigenvalues are $e^{-t\lambda_n}$ from the last line in the previous proof. Or note that

$$\int_M H(t,x,x)\,dx = \sum_n e^{-t\lambda_n} \tag{16}$$

from the previous proposition. □

3. Semigroup Theory

This summarises the results in [Eva98, Section 7.4].

3.1. **Reinterpreting the Heat Equation.** Consider the Dirichlet problem in (5) and rewrite it as

$$\begin{aligned} \frac{d}{dt}u(t) &= \Delta\, u(t), \\ u(0) &= f, \end{aligned} \tag{17}$$

which we will now interpret as an ODE in the function space L^2.

We make this more precise as follows.

First note the following properties of Δ.

(1) The domain of Δ is $\mathcal{D}(\Delta) := H_0^1 \cap H^2$, which is *dense in* L^2.
(2) Δ is *closed*. That is

$$\begin{aligned} u_k \in \mathcal{D}(\Delta),\ u_k \to u \ \&\ \Delta\, u_k \to f \text{ in } L^2, \\ \implies u \in \mathcal{D}(A) \ \&\ f = \Delta\, u. \end{aligned}$$

In fact $u_k \to u$ in H^2 under these hypotheses. To see this use the easy regularity result

$$\|u_k - u_\ell\|_{H^2} \le c\|\,\Delta\, u_k - \Delta\, u_\ell\|_{L^2} + c\|u_k - u_\ell\|_{L^2}.$$

It follows (u_k) is Cauchy in H^2. Hence $u_k \to u$ in H^2, and so $u \in \mathcal{D}(\Delta)$ and $\Delta\, u = f$.
(3) If $\lambda \ge 0$ then $\lambda - \Delta$ is invertible with domain L^2 and $\|(\lambda - \Delta)^{-1}\| \le 1/\lambda$.

The invertibility is standard and the estimate comes from the fact $(\lambda - \Delta)u = f$ implies, using the L^2 inner product and norm,

$$(\lambda u, u) + (Du, Du) = (f, u)$$

and so

$$\lambda \|u\|_{L^2} \le \|f\|\,\|u\|,$$

which gives the result.

It follows that the hypotheses of the Hille Yosida theorem are satisfied, see Theorem 3.7. This gives the existence and uniqueness of a solution

$$u(t,x) = e^{t\,\Delta} f(x), \tag{18}$$

via the semigroup $e^{t\,\Delta}$ as discussed in the following.

Regularity: Proposition 3.3 together with the fact the hypotheses of the Hille Yosida theorem are satisfied, shows that if $f \in H_0^1 \cap H^2$ then the same is true for the solution $u(t,\cdot)$ with $t > 0$.

But much more is true. In particular, if $f \in L^2$ then $u(t,\cdot)$ is C^∞ for $t > 0$. See [Ber05], in particular Proposition 3.2 (smoothing effect) p 29, Remark 3.13 p34 and Remark 2.5 p 16.

3.2. **Contractive Semigroups.** In this section X is a real Banach space, namely $L^2(\Omega)$ for us.

Definition 3.1. A family $(S(t) = S_t)_{t\ge 0}$ of bounded linear operators $S(t) \in \mathcal{B}(X)$ is a *contractive semigroup* if for $u \in X$ and $t_1, t_2 \ge 0$:

(1) $S_0 f = f$;
(2) $S_{t_1+t_2} f = S_{t_1} S_{t_2} f = S_{t_2} S_{t_1} f$;
(3) $t \mapsto S_t f$ is continuous from $[0,\infty)$ into X;
(4) $\|S_t f\| \le \|f\|$.

We can think of S_t as generating a nice flow on L^2.

But we have yet to invoke a generator A, which for us will be Δ.

Definition 3.2. We say that A is the *infinitesimal generator* of the contractive semigroup $(S_t)_{t\ge 0}$ if

$$Au = \lim_{t\to 0^+} \frac{S_t u - u}{t},$$

in the sense that the domain $\mathcal{D}(A)$ of A is the set of u for which the limit exists, and in this case Au is as shown.

It is clear that A exists and is unique.

The following differential properties are analogous to the case $S_t = e^{tA}$ where A is a linear transformation in $\mathbb{R}^n$.

Proposition 3.3. *Suppose A is the infinitesimal generator of the contractive semigroup $(S_t)_{t\geq 0}$. Assume $u \in \mathcal{D}(A)$. Then*

(1) *$S_t u \in \mathcal{D}(A)$ for $t \geq 0$.*
(2) *$AS_t u = S_t A u$ for $t \geq 0$.*
(3) *$\frac{d}{dt} S_t u = AS_t u$, and in particular $t \mapsto S_t u$ is C^1 in $(0, \infty)$.*

The proofs are straightforward, see [Eva98, p415].

In analogy with the finite dimensional case we write

$$S_t = e^{tA}. \tag{19}$$

Proposition 3.4. *Suppose A is the infinitesimal generator of the contractive semigroup $(S_t)_{t\geq 0}$. Then $\mathcal{D}(A)$ is dense in X and A is closed.*

Proof. Fix $f \in X$ and let[2]

$$u_t = \int_0^t S_s f \, ds.$$

It is straightforward to check by the semigroup properties that $u_t \in \mathcal{D}(A)$ and $Au_t = S_t f - f$. Using $u_t/t \to f$ as $t \to 0^+$, density follows.

To prove closedness suppose $u_k \to u$ and $Au_k \to f$. By Proposition 3.3

$$S_t u_k - u_k = \int_0^t S_s A u_k \, ds.$$

Letting $k \to \infty$,

$$S_t u - u = \int_0^t S_s f \, ds.$$

Hence

$$\lim_{t\to 0^+} \frac{S_t u - u}{t} = f,$$

and so $u \in \mathcal{D}(A)$ and $f = Au$. □

3.3. **Resolvents.**

Definition 3.5. The *resolvent set* $\rho(A)$ is the set of real numbers λ such that $\lambda - A : \mathcal{D}(A) \to X$ is one to one and onto. Equivalently, the *resolvent operator*

$$R_\lambda := (\lambda - A)^{-1} : X \to X \tag{20}$$

is well defined.

It is straightforward to check that R_λ is closed and hence bounded by the closed graph theorem.

Proposition 3.6. [3]

Suppose $\lambda, \mu \in \rho(A)$. Then:

(1) *$AR_\lambda u = R_\lambda A u$ for $u \in \mathcal{D}(A)$.*
(2) *$R_\lambda - R_\mu = (\mu - \lambda) R_\mu R_\lambda$ (resolvent identity).*

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[2] Here we are using integration of Banach space valued functions, see [Eva98, pp649,670].

[3] For a "bare hands" approach to the last claim in the case of the heat equation we follow [Eva98, Example 4, §4.3.2].

The *Laplace transform* of $u \in L^1[0, \infty)$ is

$$u^\#(\lambda, t) := \int_0^\infty e^{-\lambda t} u(t) \, dt \text{ for } \lambda \geq 0.$$

Suppose

$$\begin{aligned} \partial_t u(t, x) &= \Delta \, u(t, x) \quad t > 0, \ x \in \Omega, \\ u(0, x) &= f(x). \end{aligned}$$

Then

$$\Delta_x \, u^\#(\lambda, x) = \int_0^\infty e^{-\lambda t} \partial_t u(t, x) \, dt = \lambda \int_0^\infty e^{-\lambda t} u(t, x) \, dt - f(x).$$

That is, *the solution of the resolvent equation*

$$(\lambda - \Delta) v(x) = f(x),$$

is the Laplace transform of the solution of the original problem with initial data f.

(3) $R_\lambda - R_\mu = R_\mu R_\lambda$.
(4) $R_\lambda f = \int_0^\infty e^{-\lambda t} S_t f\, dt$ *for* $f \in X$ *(the resolvent operator is the Laplace transform of the semigroup). It follows* $\|R_\lambda\| \le \lambda^{-1}$.

Proof. The first 3 parts are straightforward computations, see [Eva98, p417].

For the last part let

$$\widetilde{R}_\lambda f = \int_0^\infty e^{-\lambda t} S_t f\, dt.$$

The integral is well defined as $\|S_t\| \le 1$. First we show

$$(\lambda - A)\widetilde{R}_\lambda f = f, \tag{21}$$

$$\text{i.e.} \quad A\widetilde{R}_\lambda f = -f + \lambda \widetilde{R}_\lambda f,$$

$$\text{i.e.} \quad \lim_{h\to 0^+} \frac{S_h \widetilde{R}_\lambda f - \widetilde{R}_\lambda f}{h} = -f + \lambda \widetilde{R}_\lambda f.$$

This is checked by direct computation.

But $A\widetilde{R}_\lambda u = \widetilde{R}_\lambda A u$ for $a \in \mathcal{D}(A)$ from the closedness of A. It follows

$$\widetilde{R}_\lambda(\lambda - A)u = u \tag{22}$$

for $u \in \mathcal{D}(A)$.

From (21) and (22) it follows that $\widetilde{R}_\lambda = (\lambda - A)^{-1} = R_\lambda$. □

3.4. Generating contractive semigroups.

Theorem 3.7 (Hille-Yosida theorem)**.** *Let A be a closed and densely defined operator on the Banach space X. Then A is the generator of a contraction semigroup iff*

$$\lambda - A : \mathcal{D} \to X \text{ is one-one and onto, i.e. } \lambda \in \rho(A),$$

$$\|R_\lambda\| \le \frac{1}{\lambda} \text{ for } \lambda > 0.$$

Proof Outline. Let

$$A_\lambda = -\lambda + \lambda^2 R_\lambda = \lambda A R_\lambda = A(I - \lambda^{-1}A)^{-1}.$$

One first checks that

$$A_\lambda u \to Au \text{ as } \lambda \to \infty, \text{ for } u \in \mathcal{D}(A).$$

Next let

$$S_\lambda(t) := e^{tA_\lambda} = e^{-\lambda t} \sum_{k\ge 0} \frac{(\lambda^2 t)^k}{k!} R_\lambda^k.$$

Then $\|S_\lambda(t)\| \le 1$ essentially since $\|R_\lambda\| \le \lambda^{-1}$.

Using

$$\frac{d}{dt} S_\lambda(t)u = A_\lambda S_\lambda(t)u = S_\lambda(t) A_\lambda u,$$

it follows that

$$\|S_\lambda(t)u - S_\mu(t)u\| \le t\|A_\lambda u - A_\mu u\| \to 0 \text{ as } \lambda, \mu \to \infty.$$

Hence $S(t)u := \lim_{\lambda\to\infty} u$ exists for $t \ge 0$ and $u \in \mathcal{D}(A)$.

It also follows that $S(t)$ is a contraction semigroup.

By a limiting argument, one shows that A is the generator of $S(t)$. □

4. Weyl's Formula

4.1. The Formula and Consequences.

Theorem 4.1. *For a compact d-dimensional Riemannian manifold M,*[4]

$$\sum_n e^{-t\lambda_n} = \int_M H(t,x,x)\, dx \sim \frac{\mathrm{vol}(M)}{(4\pi)^{d/2}} t^{-d/2} \text{ as } t \to 0^+. \tag{23}$$

[4] $f(t) \sim g(t)$ means $\lim_t f(t)/g(t) = 1$, where $t \to 0$ or $t \to \infty$ etc. as will be clear from context.

Proof. The equality follows from (16). The asymptotic equivalence follows from (31). Using this,

$$\int_M H(t,x,x)\,dx \sim \int_M \frac{1}{(4\pi t)^{d/2}} = \frac{\mathrm{vol}(M)}{(4\pi t)^{d/2}}.$$

□

The point is that *the spectrum of the Laplacian, or equivalently by* (16) *the short term asymptotics of the heat kernel, determine the dimension and (more surprisingly) the volume of the manifold.* (In fact much more geometry is determined from this data.)

Let $N(\lambda)$ be the number of eigenvalues $\le \lambda$, or equivalently let dN denote the discrete measure with unit mass at each λ_n. Then we can write

$$\sum_n e^{-t\lambda_n} = \int e^{-\lambda t}\,dN(\lambda).$$

By the Tauberian theorem 5.2, it then follows that (23) is equivalent to

$$N(\lambda) \sim \frac{\mathrm{vol}(M)}{(4\pi)^{d/2}\,\Gamma\left(\frac{d}{2}+1\right)}\,\lambda^{d/2} \text{ as } \lambda\to\infty. \tag{24}$$

This is clearly equivalent to

$$\frac{\lambda_n^{d/2}}{n} \sim \frac{(4\pi)^{d/2}\,\Gamma\left(\frac{d}{2}+1\right)}{\mathrm{vol}(M)} \text{ as } n\to\infty. \tag{25}$$

Note that

$$\Gamma\left(\frac{d}{2}+1\right) = \begin{cases} \left(\frac{d}{2}\right)! & d \text{ even},\\ \left(\frac{d}{2}-1\right)\left(\frac{d}{2}-2\right)\cdot\ldots\cdot\frac{1}{2}\sqrt{\pi} & d \text{ odd}.\end{cases}$$

4.2. **Construction of a Parametrix.** This is the first step in the construction of the heat kernel. We outline the construction following [Ros97, Section 3.2, pp 90–99] and [Bas08].[5]

The heat kernel for $\mathbb{R}^n$ is

$$\frac{1}{(4\pi t)^{n/2}} e^{-|x-y|^2/4t}.$$

Thus a reasonable first approximation to the heat kernel on M for x near y and t small is

$$\widehat{K}(t,x,y) = \frac{1}{(4\pi t)^{n/2}} e^{-d(x,y)^2/4t} \tag{26}$$

where d is the geodesic distance on M. Note that $\widehat{K}(0,x,y) = \delta_x(y)$.

In order to build this into the heat kernel by iterating Duhamel's construction, we first need need to replace $\widehat{K}(t,x,y)$ by $K_0(t,x,y)$ where K_0 is still a delta function at $t=0$ but where also $(\partial_t - \Delta_x)K_0(t,x,y) \to 0$ as $t\to 0$.

For this assume we can write for each $k\ge 1$

$$K_0(t,x,y) = \widehat{K}(t,x,y)\big(1 + tu_1(x,y) + \cdots + t^k u_k(x,y)\big) \tag{27}$$

By iteratively solving PDE's (in fact ODE's for $d = d(x,y)$ in

$$U_\epsilon = \{(x,y)\in M\times M : (x,y) < \epsilon\}$$

for $\epsilon > 0$ sufficiently small we successively find C^∞ functions u_i on U_ϵ such that

$$(\partial_t - \Delta_x)K_0(t,x,y) = -\widehat{K}(t,x,y)t^k\,\Delta\,u_k.$$

Choosing $k > n/2$ and a cut-off function equal to 1 on $U_{\epsilon/2}$ and 0 on $M\times M\setminus U_\epsilon$, one gets

$$K_0(t,x,y) \in C^\infty\big((0,\infty)\times M\times M\big),$$

$$\lim_{t\to 0^+}\int_M K_0(t,x,y)\,f(y)\,dy = f(x) \quad \forall f\in C^0(M), \quad \text{i.e. } K_0(0,x,y) = \delta_y(x), \tag{28}$$

$$Q(t,x,y) = O(t) \text{ as } t\to 0^+, \quad Q(t,x,y) := (\partial_t - \Delta_x)K_0(t,x,y).$$

[5]There is a problem with the discussion in [Bas08]. He effectively works with $\widehat{K}$ instead of K_0. But since $(\partial_t - \Delta_x)\widehat{K}(t,x,y)$ blows up as $t\to 0^+$, one needs instead to work with K_0.

The treatment in [Gri04] avoids this two step process and in this way also gives an introduction to pseudo differential operators.

In view of (28) we say that K_0 is a *parametrix*.

4.3. **Construction of the Heat Kernel and Consequences.** Motivated by the argument in (13) we define

$$K_1(t,x,y) = K_0(t,x,y) - \int_0^t \int_M K_0(t-s,x,z)\, Q(s,z,y)\, dz\, ds,$$
$$\text{i.e. } K_1 =: K_0 - K_0 \# Q.$$

Thus in the second term on the right we are using the kernel K_0 to form a weighted "sum" of the Q over $[0,t]\times M$, essentially just taking a convolution in time and space in a natural manner. The point is that as in (13) we get

$$\begin{aligned}(\partial_t - \Delta_x)K_1(t,x,y) &= (\partial_t - \Delta_x)K_0(t,x,y) - \int_M K_0(0,x,z)\, Q(t,z,y)\, dz \\ &\quad - \int_0^t \int_M (\partial_t - \Delta_x)K_0(t-s,x,z)\, Q(s,z,y)\, dz\, ds \\ &= Q(t,x,y) - Q(t,x,y) - Q\#Q(t,x,y) = -Q\#Q(t,x,y).\end{aligned}$$
$$\text{i.e. } (\partial_t - \Delta_x)K_1 = -Q\#Q.$$

For the penultimate line we used in order the definition of Q in the last line of (28), the fact $K_0(0,x,y)$ is a delta "function" from the second line in (28), and the definition of Q together with the definition of $\#$.

Similarly, setting

$$K_2 = K_1 + K_0\#Q\#Q,$$

one gets

$$(\partial_t - \Delta_x)K_2 = -Q\#Q + Q\#Q + Q\#Q\#Q = Q\#Q\#Q.$$

In general

$$K_k = K_0 - K_0\#Q + K_0\#Q\#Q - K_0\#Q\#Q\#Q + \cdots + (-1)^k K_0 \#^k Q, \tag{29}$$
$$(\partial_t - \Delta_x)K_k = \#^k Q.$$

Each time we take another "power" of Q we pick up another factor of t, and in fact because of the integration we pick up $t^k/k!$ with K_k. It follows that the heat kernel

$$H(t,x,y) := \lim_{k\to\infty} K_k(t,x,y) \tag{30}$$

exists for $t>0$ and $x,y\in M$, and similarly for all derivatives.

In particular,

$$\begin{aligned}H(t,x,y) &= K_0(t,x,y)\big(1+O(t)\big) \quad \text{as } t\to 0 \\ &= \frac{1}{(4\pi t)^{n/2}} e^{-d(x,y)^2/4t}\big(1+O(t)\big) \quad \text{as } t\to 0\end{aligned} \tag{31}$$

by (27) and (26).

Moreover,

$$(\partial_t - \Delta_x)H(t,x,y) = \lim_{k\to\infty} \#^k Q(t,x,y) = 0. \tag{32}$$

Also

$$H(0,x,y) = K_0(0,x,y) = \delta_y(x). \tag{33}$$

From the last three equations it follows that H is the heat kernel on M.

5. Karamata's Tauberian Theorem

Suppose μ is a Borel measure on $[0,\infty)$. Provided

$$(\mathcal{L}\mu)(t) := \int_0^\infty e^{-tx} d\mu(x) < \infty \tag{34}$$

for all $t>0$, then $\mathcal{L}\mu$ is called the *Laplace Transform* of μ.

Note that μ may grow at any polynomial rate at ∞.

Remark 5.1.

(1) In the following, the easy (Abelian) direction is (35) implies (36). The reverse (Tauberian) direction is more difficult.
(2) For $\alpha > 0$, $\Gamma(\alpha) := \int_0^\infty x^{\alpha-1}e^{-x}\,dx$, $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$, $\Gamma(n+1) = n!$ if $n \in \mathbb{N}_0$. Also, $\Gamma(1/2) = \sqrt{\pi}$.
(3) Setting $\mu = \sum_{n\ge 0} a_n\delta_n$, $e^{-\lambda} = r$, $\alpha = 0$ gives the usual Abel summation result. Except that we are using positive measures and hence $a_n \ge 0$. However, once can also get results for signed measures, see [Kor04, pp 31–33].
    In fact, the usual Abelian and Tauberian theorem, with a general one sided assumption (Tauberian condition) in the latter case, also follow. See [Kor04, §§1.15, 1.21].
(4) One does need a Tauberian condition, which here is that the measures are nonnegative. For example, if $\mu = \sum_{n\ge 0}(-1)^n\delta_n$ and $\alpha = 0$ then
$$\lim_{t\to 0}\int e^{-tx}d\mu(x) = \sum_{n\ge 0} e^{-tx}(-1)^n = \frac{1}{1+e^{-t}} \to \frac{1}{2},$$
but
$$\lim_{x\to\infty}\mu[0,x] = \sum_{n\ge 0}(-1)^n\delta_n$$
does not exist.
(5) The proof is from [Sim79, pp 108 – 109]. See also [Tay96, pp 89, 90] (a confusing error: in the proof there, b should be replaced by $a/\Gamma(\alpha)$) and [Are05, Section 6.2].
(6) For Weiner's Tauberian Theorem see [Rud91, p 229] and for his general approach [Kor04, Chapter 2]. For a survey which discusses both Karamata and Weiner, and how their work gives the Hardy Littlewood results, see [Bor05].

□

Theorem 5.2. *Suppose $\alpha \ge 0$ and μ is as above.*[6] *Then each of the following implies the other:*

$$\mu[0,x] \sim \frac{ax^\alpha}{\Gamma(\alpha+1)} \text{ as } x\to\infty, \tag{35}$$

$$\int e^{-tx}d\mu(x) \sim \frac{a}{t^\alpha} \text{ as } t \to 0^+. \tag{36}$$

(*The Tauberian theorem is* (36) $\Longrightarrow$ (35) *and the Abelian theorem is* (35) $\Longrightarrow$ (36).)

Proof. If $\alpha = 0$ then (36) says $\mu[0,\infty) = a$ and (35) says $\int e^{-tx}d\mu(x) \to a$ as $t \to 0+$. Both are equivalent by the monotone convergence theorem.

Assume (36) and $\alpha > 0$.
Let $\mu_t = t^\alpha\rho_{t\#}\mu$. Then for $s > 0$,
$$\begin{aligned}\int e^{-sx}d\mu_t(x) &= t^\alpha\int e^{-stx}d\mu(x) \quad \text{by the defn of } \mu_t\\ &\to as^{-\alpha} \text{ as } t\to 0^+ \quad \text{from (36)}\\ &= \frac{a}{\Gamma(\alpha)}\int e^{-sx}x^{\alpha-1}dx \quad \text{by the defn of } \Gamma.\end{aligned}$$
By Stone Wierstrass (e.g. as in [Con90]) since the algebra generated by e^{-sx} for $s > 0$ is dense in $C_0(\mathbb{R}^+)$,
$$\mu_t \rightharpoonup \frac{a}{\Gamma(\alpha)}x^{\alpha-1}L^1 \text{ as } t\to 0^+.$$
Hence as $t \to 0^+$,
$$t^\alpha\mu[0,t^{-1}] = \mu_t[0,1] \to \frac{a}{\Gamma(\alpha)}\int_0^1 x^{\alpha-1}dx = \frac{a}{\alpha\Gamma(\alpha)} = \frac{a}{\Gamma(\alpha+1)}.$$
Replacing t by x^{-1} gives (35).

Assume (35) and $\alpha > 0$.

[6] Either of the two following conditions implies finiteness as in (34).

Suppose $\epsilon > 0$. Then

$$\left|\mu[0,x] - \frac{ax^\alpha}{\Gamma(\alpha+1)}\right| \le \frac{\epsilon}{\Gamma(\alpha+1)} x^\alpha \quad \text{if } x \ge N(\epsilon).$$

Since

$$\frac{1}{\Gamma(\alpha+1)} \int e^{-tx} dx^\alpha = t^{-\alpha}$$

from the definition of the Gamma function, and since

$$\int_0^{N(\epsilon)} e^{-tx} d\mu(x) \le \mathbb{M}(\mu),$$

it follows that

$$\left| \int e^{-tx} d\mu(x) - at^{-\alpha} \right| \le \epsilon t^{-\alpha} + \mathbb{M}(\mu).$$

Since $\epsilon > 0$ is arbitrary,

$$\left| t^\alpha \int e^{-tx} d\mu(x) - a \right| \le t^\alpha \mathbb{M}(\mu).$$

This gives (36) with a control on the error term. □

6. Finite Dimensional Background: Kernels and Trace

An $n \times n$ matrix is *Gramian* if it is of the form $A_{ij} = \langle v^i, v^j \rangle$ for vectors $v^1, \dots v^n \in \mathbb{R}^n$. It is clearly non negative and symmetric. The converse is true.

Proposition 6.1. *A non negative symmetric matrix is Gramian.*

Proof. By symmetry there is an orthonormal change of basis U so that $U^{-1}AU$ is diagonal. By positive semidefiniteness, all entries on the diagonal are non negative.

So $A^{1/2}$ exists (square root of the diagonal matrix in the new coordinates) and is also symmetric positive, semidefinite, and so

$$A_{ij} = \langle Ae_i, e_j \rangle = \langle A^{1/2}A^{1/2}e_i, e_j \rangle = \langle A^{1/2}e_i, A^{1/2}e_j \rangle. \tag{37}$$

Then A is Gramian with $v^i = A^{1/2}e_i$. □

Remark 6.2. Equivalently, let ϕ^k be the e-value of A with e-value λ_k. Then setting $A^{1/2}\phi^k := \lambda_k^{1/2}\phi^k$ for all k, clearly defines the square root of A. □

Remark 6.3 (Kernel Perspective). This is useful for comparison with integral operators and their kernels.

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ correspond to the matrix A.
Think of $u = (u_1, \dots, u_n) \in \mathbb{R}^n$ as a function on $X = \{1, \dots, n\}$.
Let μ be the measure $\sum_{j=1}^n \delta_j$.
Let $K(i,j) = A_{ij}$ be the "kernel" of T.
Then

$$(Tu)(i) = \sum_{j=1}^n A_{ij}u(j) = \int_X K(i,j)\, u(j)\, d\mu. \tag{38}$$

□

For a general integral operator, the kernel K is the continuous matrix representative. In case K is *symmetric*, the bilinear expansion of K corresponds to diagonalisation of the matrix by means of an orthonormal basis, see Proposition 8.5.

Remark 6.4 (Kernel Decomposition). Since (ϕ_k) form an orthonormal basis, using (37),

$$A_{ij} = \langle Ae_i, e_j \rangle = \left\langle A\left(\sum_k \phi_i^k \phi^k\right), \sum_k \phi_j^k \phi^k \right\rangle = \left\langle \sum_k \lambda_k \phi_i^k \phi^k, \sum_k \phi_j^k \phi^k \right\rangle = \sum_k \lambda_k \phi_i^k \phi_j^k.$$

That is, in functional and kernel notation,

$$K(i,j) = \sum_k \lambda_k\, \phi^k(i)\, \phi^k(j).$$

Note that this gives another Gramian representation via

$$A_{ij} = \langle \theta^i, \theta^j \rangle, \quad \text{where } \theta^i := \sum_k \lambda_k^{1/2} \phi_i^k e_k.$$

Compare this with

$$A_{ij} = \langle v^i, v^j \rangle, \quad \text{where } v^i = A^{1/2} e_i = A^{1/2} \sum_k \phi_i^k \phi^k = \sum_k \lambda_k^{1/2} \phi_i^k \phi^k.$$

□

Proposition 6.5 (Trace Formula)**.** *The trace[7] of $T : \mathbb{R}^n \to \mathbb{R}^n$ is invariant under change of coordinates.*

Moreover

$$\operatorname{Tr}(T) = \sum_{i=1}^n \lambda_i(T), \tag{39}$$

the sum of the eigenvalues of T according to their algebraic multiplicity.

Proof. Since $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$ it follows $\operatorname{Tr}(S^{-1}TS) = \operatorname{Tr}(SS^{-1}T) = \operatorname{Tr} T$, giving the first claim. The second follows by choosing a basis so that T is in Jordan normal form. □

Remark 6.6. A coordinate free definition is to note $\operatorname{Hom}(V, V) = V \otimes V^*$, define $\operatorname{Tr}(v \otimes f) = \langle v, f \rangle$, and extend by linearity. One still needs to check the definition is consistent

Proposition 6.7 (Polar decomposition)**.** *Suppose $T : \mathbb{R}^n \to \mathbb{R}^n$. Then $T = UA$ where*

(1) *$A = (T^*T)^{1/2}$ is non negative symmetric is called the absolute value of T,*
(2) *U is an isometry on $\mathcal{R}(A)$, $U = 0$ on $\mathcal{R}(A)^\perp$.*

Proof. Straightforward, see [Lax02, pp 330-1]. Define $A = (T^*T)^{1/2}$ as in (1) in the proposition, define $U(Au) = Tu$ on the range of A, and define $U = 0$ on $\mathcal{R}(A)^\perp$. □

Remark 6.8. The non zero e-values of A are called the *singular values* of T and written $s_j = s_j(T)$.

We can define Tr on a finite dimensional space with inner product space $(\cdot)$ by

$$\operatorname{Tr}(T) = \sum_i (Te_i, e_i), \tag{40}$$

for any orthonormal basis e_i. It is easily checked this is independent of the choice of orthonormal basis and agrees with the previous definition.

If f_i is an orthonormal basis of e-vectors for A with the e-values $s_i(T)$, which is possible since A is symmetric, then clearly

$$\operatorname{Tr}(T) = \sum_i s_i(T)(Uf_i, f_i) \tag{41}$$

and so

$$|\operatorname{Tr}(T)| \le \sum_i s_i(T). \tag{42}$$

Remark 6.9. The *trace norm* is defined by

$$\|\operatorname{Tr}(T)\|_{tr} = \sum_i s_i(T). \tag{43}$$

This *is* a norm and $\|\operatorname{Tr}(T)\|_{tr} = \|\operatorname{Tr}(T^*)\|_{tr}$.

The proof is straightforward, see [Lax02, Theorem 2 p 331], where a more general result in Hilbert spaces is proved. See also the following section here.

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[7] The *trace* is the sum of the diagonal elements.

7. Trace Class

In the following, $\mathcal{H}$ is a separable Hilbert space over $\mathbb{C}$, and $T : \mathcal{H} \to \mathcal{H}$ is linear and compact.

Theorem 7.1. *A compact linear map $T : \mathcal{H} \to \mathcal{H}$ can be written as*

$$T = UA \tag{44}$$

where

(1) *$A = (T^*T)^{1/2}$ is non negative compact symmetric and is called the <u>absolute value</u> of T,*
(2) *U is an isometry on $\overline{\mathcal{R}(A)}$, $U = 0$ on $\mathcal{R}(A)^\perp$.*

Proof. Essentially as for Proposition 6.7. In fact, the result is true for bounded operators. □

Remark 7.2. Equivalently, T is of the form

$$Tf = \sum_k s_k \langle f, v_k \rangle u_k,$$

where $(u_k)_{k\geq 1}$ and $(v_k)_{k\geq 1}$ are orthonormal bases and $0 \leq \alpha_k \to 0$ as $k \to \infty$. The s_k are the *singular values* of T and the e-values of A, and are written in nondecreasing order. □

Remark 7.3. Recall that a *compact <u>symmetric</u>* T has an orthonormal basis ϕ_k of e-vectors with e-vectors λ_k such that $\lambda_k \to 0$ as $k \to \infty$. Moreover, any T of this form is compact and symmetric. See[SS05, pp 190–193].

The e-values and the singular values are then equal up to sign. □

Remark 7.4 (Overview). (See the following discussions and[Wikb].) With the previous decomposition we have:

$$T \text{ compact} \iff s_k \to 0$$
$$T \text{ is Hilbert Schmidt} \iff \|T\|_2 := \sum_k s_k^2 < \infty$$
$$T \text{ is trace class} \iff \|T\| := \sum_k s_k < \infty$$
$$T \text{ is finite rank} \iff s_k = 0 \text{ eventually.}$$

In particular,

$$\{\text{finite rank}\} \subset \{\text{trace class}\} \subset \{\text{Hilbert Schmidt}\} \subset \{\text{compact}\}.$$

Also,

$$\|T\| \leq \|T\|_2 \leq \|T\|_1.$$

□

Definition 7.5. The compact map T is *trace class* if the trace norm

$$\|T\|_{tr} := \|T\|_1 := \sum_i s_i(T) < \infty.$$

Proposition 7.6. *Trace class operators form a Banach space under the trace norm. Moreover*

$$\|T\|_{tr} = \sup \sum_n |(Tf_n, e_n)|, \tag{45}$$

where the sup is over all pairs of orthonormal bases (f_n) and (e_n).

Proof. See [Lax02, pp 331, 333]. Main ideas are in the previous finite dimensional section. □

Definition 7.7. The *trace* of a bounded operator is

$$\mathrm{Tr}(T) = \sum_n (Tf_n, f_n), \tag{46}$$

for any orthonormal basis (f_n), provided the series converges.

Theorem 7.8. *If T is trace class then (46) converges absolutely to a limit independent of the orthonormal basis. Moreover,*

(1) *$|\operatorname{Tr} T| \leq \|T\|_{\mathrm{Tr}}$,*
(2) *Tr is linear,*

(3) $\operatorname{Tr} T^* = \overline{\operatorname{Tr} T}$,
(4) $\operatorname{Tr}(TB) = \operatorname{Tr}(BT)$ *for any bounded operator* B.

Proof. See [Lax02, Theorems 3,4 p 333]. Essentially no new ideas above the previous section. □

The following major result is due to Lidskii, 1959.

Theorem 7.9 (Trace Formula). *If T is trace class then*

$$\operatorname{Tr} T = \sum_j \lambda_j(T), \tag{47}$$

summing according to the algebraic multiplicity.

Proof. See [Lax02, pp334–341].

If T is normal then there is an orthonormal basis of e-vectors and the result follows immediately from the definition of Tr.

In general, first recall that the *algebraic multiplicity* of an eigenvalue λ is the dimension of the corresponding space of its generalised e-vectors, i.e. the union of the null spaces of $(\lambda I - T)^i$ over all positive integers i. It is finite for compact T. See [Lax02, p267].

Using Gram Schmidt we can get an orthonormal basis of each generalised e-space. Summing over all such basis vectors would give the result if they span $\mathcal{H}$, but this need not be so. In general,

$$\operatorname{Tr} T = \sum_i \lambda_i + \sum_j (Th_j, h_j)$$

where the h_j form an orthonormal basis for the complement space.

The difficult part is to show the second term is zero. See [Lax02, pp 335–341]. □

8. Hilbert Schmidt Integral Operators

*** Note discussion in [Dod81].

Definition 8.1. Suppose $E \subset \mathbb{R}^n$ is measurable. If (the *kernel*) $K : E \times E \to \mathbb{R}$ satisfies $\int_{E\times E} K^2(x,y) < \infty$ then $T : L^2(E) \to L^2(E)$ given by

$$(Tf)(x) = \int_E K(x,y)\, f(y)\, dy$$

is called a *Hilbert Schmidt integral operator.*

In practice, almost all bounded operators of interest are integral operators in one or several dimensions. [Lax02, p 342].

Remark 8.2. We just consider Hilbert Schmidt integral operators on $\mathcal{H} = L^2(E)$ as above. For such function spaces this is the same as being Hilbert Schmidt in the sense of Remark 7.4. See for example [HL99, Q21, pp 140, 141, with Hints].

See [Lax02, p352] for a summary of the properties of general Hilbert Schmidt operators. □

Proposition 8.3.

$$\|T\| \le \|T\|_2 = \|K\|_{L^2(E\times E)}$$
$$T^* \text{ has kernel } \overline{K}$$
$$\text{Hilbert Schmidt} \implies \text{compact}$$

Proof. Straightforward. See [SS05, pp 187, 190]. □

Proposition 8.4. *If $E = B_1(0) \subset \mathbb{R}^n$ and $|K(x,y)| \le c|x-y|^{-d+\alpha}$ for some $\alpha > 0$, then T is bounded and compact. T is HS iff $\alpha > d/2$.*

Proof. See [SS05, Q28, p199, with Hint]. Straightforward. □

Proposition 8.5. *Suppose K is a real symmetric HS kernel. Then $T : L^2 \to L^2$ is symmetric (and compact as noted before).*

Let ϕ_k be the corresponding e-functions with e-values λ_k. Then

(1) $\sum_k \lambda_k^2 < \infty$,
(2) $K(x,y) = \sum_k \lambda_k \phi_k(x)\, \phi_k(y)$ *(L^2 convergence),*

(3) *If we just assume T is symmetric and compact, then T is HS iff $\sum_k \lambda_k^2 < \infty$.*

Proof. Straightforward. See [SS05, Q34 p201, with Hint]. For the second claim use the right side as a kernel and show it has the same effect on each ϕ_k, namely $\lambda_k \phi_k$. □

9. **Mercer's Theorem**

See [Lax02, pp 344], [RSN90, pp245, 246] and [Wika].

*** Note discussion in [Dod81].

In the following, think of the case $E = \overline{\Omega}$ where Ω is a bounded open subset of $\mathbb{R}^n$, or E is a fractal.

By K is non negative definite is meant that $(Ku, u) \geq 0$ for all $u \in L^2(E)$. Or equivalently by approximation arguments, $\sum_{i,j} K(x_i, x_j)c_i c_j \geq 0$ for all finite sequences $x_1, \dots, x_n \in E$ and $c_1, \dots, c_n \in \mathbb{R}$.

Theorem 9.1. *Suppose E is a compact Hausdorff space and μ is a Radon measure on X. Suppose $K(x, y)$ is continuous, symmetric, and non negative definite. Let T be the corresponding integral operator (as in Definition 8.1).*

From Proposition 8.5, $T : L^2(X) \to L^2(X)$ is compact in the L^2 norm. Let ϕ_k be the corresponding e-functions with e-values λ_k.

(1) *The e-functions are continuous,*
(2) *$T : C(X) \to C(X)$ is non negative, symmetric, and compact in the sup norm,*
(3) *$K(x, y) = \sum_k \lambda_k\, \phi_k(x)\, \phi_k(y)$, where the convergence is absolute and uniform.*
(4) *$\int K(x, x)\, dx = \sum_j \lambda_j = Tr(T)$. In particular, T is trace class.*

Proof. (See [Lax02, pp 343, 344] for details.)

It is easy to check Tf is continuous for $f \in L^2$. It follows that the e-functions are continuous.

$K(x, x) \geq 0$, from the non negative property of K and its continuity.

T is non negative and symmetric from Proposition 8.5. Compactness in the sup norm follows from the Arzela Ascoli theorem.

The series $\sum_k \lambda_k \phi_k(x)\, \phi_k(x)$ is an increasing series of positive continuous functions bounded by $K(x, x)$. By Dini's theorem this gives uniform convergence.

Denote the kernel corresponding to the limit by K_∞. It follows from looking at the e-functions that $K = K_\infty$.

Finally, set $x = y$ in (3) and integrate to get (4). □

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